Private cloud creation

Private clouds are created to manage the IT requirements and utilisation of computing resources on premises. They use virtualisation software like VMware, virtual box, Zen, and others to manage the resources. A private cloud helps in cutting down costs and improving performance if hardware is sized for the peak requirements of a company. These clouds can be created in half an hour to two hours.

A virtual private cloud is created for a company that has different computing requirements on a public cloud. Such a cloud is isolated from another company's users, and its resources are private. A hosted private cloud is one where the resources are not accessible and shared with other companies. The hardware is maintained by the company. In a managed private cloud, the service provider manages the storage and identity management services. An on-premises private cloud hosts the cloud infrastructure in the corporate network.

Autoscaling

Private clouds can be autoscaled using policies. Policy has terms such as memory, disk space, processing power, CPU, utilisation, etc. A private cloud virtualisation manager checks these terms and manages the cloud by adding and deleting nodes. The thresholds on the node count can be set in the virtualisation manager policy. Averages of the limits of the usage can be collected and also configured in the policy.

Policy has an option for choice of metrics related to processing power, disk space, and memory. Virtualisation managers can provide options to change the node group size, based on disk space limits. Nodes are added when the space utilisation crosses the limit.

> If someone asks me what cloud computing is, I try not to get bogged down with definitions. I tell them that, simply put, cloud computing is a better way to run your business.
>
> *— **Marc Benioff,** founder, CEO and chairman, Salesforce*

Resource management

In the private cloud, resources are managed by the virtualisation manager. A node group can be configured with varying sizes. The policies can be created for resource monitoring and usage. Elasticity options can be managed by the virtualisation manager policy. The management dashboard provides information like node count, location, DNS servers, node groups, and operational state.

Private clouds can be deleted by deleting the node groups and the nodes within these groups. The data can be backed up to the on-premises backup archive.

The access control for users can be changed for configuring the identities, managing users, port group deletion, and service account creation. The admin has the access control to manage the private cloud. Roles can be created on the private cloud by adding the identity provider. Groups can be created to manage the access for a group of users.

Private cloud networking

Cloud virtual private networking can be used for site-to-site connection between the cloud and the on-premises servers. The virtualisation manager is configured on the isolated hardware nodes on the cloud. Single points of failures are avoided on the private cloud. Node groups on the private cloud have high availability and resilience options. Private clouds can be shielded against network outages by using vSAN.

What's next?

There are a lot of private cloud vendors in the market as of September 2021. Hewlett Packard Enterprise has the Helion cloud suite, cloud system hardware, managed private cloud, and managed virtual private cloud software in its 2021 product release. VMware has vSphere, vRealize cloud management, and cloud foundation software defined data centre management platforms. Dell EMC provides cloud management and cloud security features. Oracle platform has compute and storage features. IBM offers managed services, cloud security tools, cloud management and orchestration modules. Red Hat provides a cloud suite, OpenStack and Gluster storage software modules.

These private clouds have many useful features and are becoming popular in the enterprise space. About 50 per cent of the CXOs queried in a survey conducted by Deloitte in 2021 stated that private clouds are more cost-effective than public clouds. That is the reason why they have a major share in the global cloud market today. END

By: Bhagvan Kommadi

'Quantica Computacao', India's first quantum computing startup, was founded by the author. He has 20 years of experience in product development and IT services.